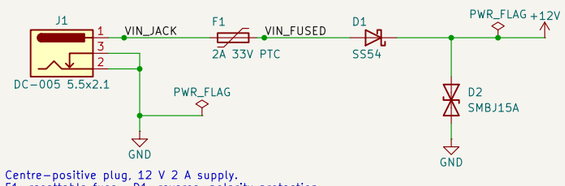
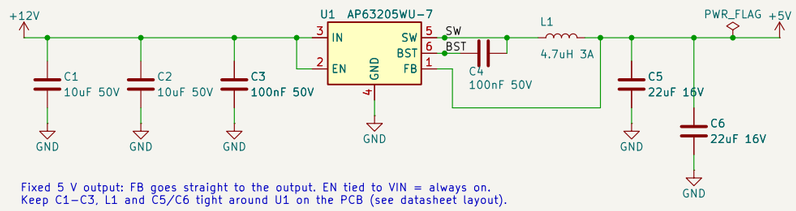
1 · POWER INPUT 12 V DC



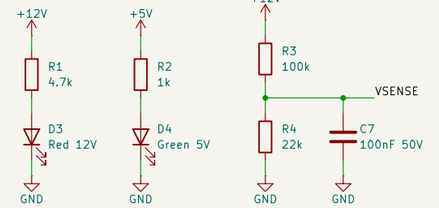
Centre-positive plug, 12 V 2 A supply.
F1: resettable fuse. D1: reverse-polarity protection.
D2: clamps plug-in voltage spikes.

2 · 5 V / 2 A BUCK (AP63205)



Fixed 5 V output: FB goes straight to the output. EN tied to VIN = always on.
Keep C1–C3, L1 and C5/C6 tight around U1 on the PCB (see datasheet layout).

3 · POWER LEDS + 12 V SENSE



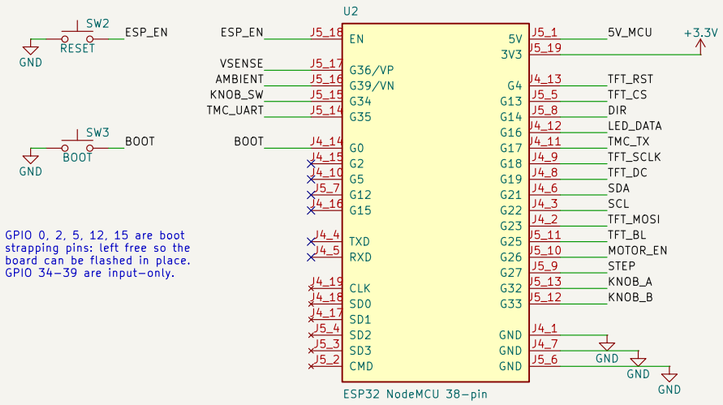
12 V -> 2.16 V at GPIO36
(15 V -> 2.70 V)

4 · TEST POINTS

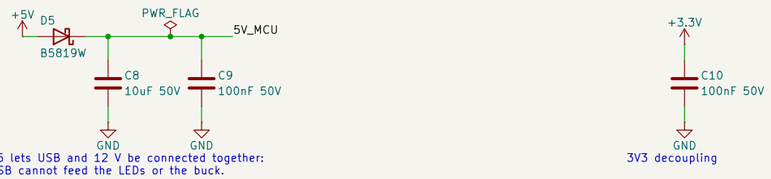


TP5 = output of the last LED
(to chain more LEDs later).

5 · ESP32 NODEMCU (in sockets)



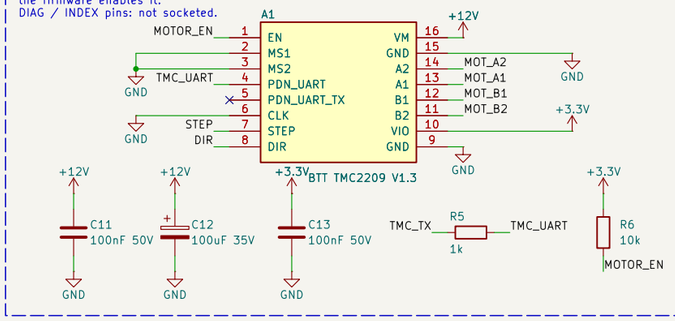
GPIO 0, 2, 5, 12, 15 are boot strapping pins: left free so the board can be flashed in place. GPIO 34–39 are input-only.



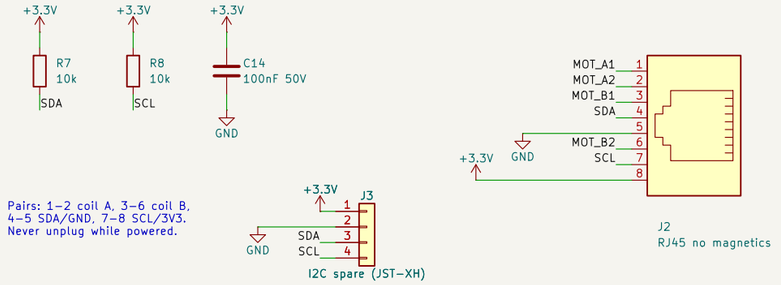
D5 lets USB and 12 V be connected together:
USB cannot feed the LEDs or the buck.

6 · STEPPER DRIVER (BTT TMC2209 V1.3, in sockets)

MS1 = MS2 = GND: UART address 0.
R6 keeps the driver off until
the firmware enables it.
DIAG / INDEX pins: not socketed.

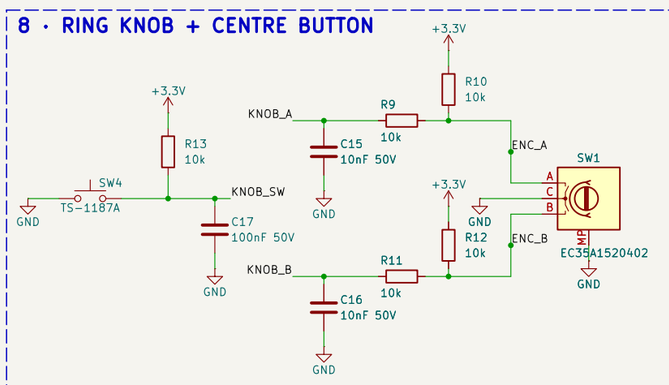


7 · MOTOR + ENCODER LINK (RJ45) – NOT ETHERNET



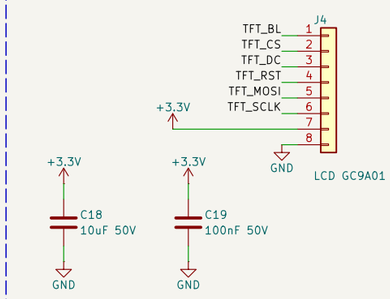
Pairs: 1–2 coil A, 3–6 coil B,
4–5 SDA/GND, 7–8 SCL/3V3.
Never unplug while powered.

8 · RING KNOB + CENTRE BUTTON

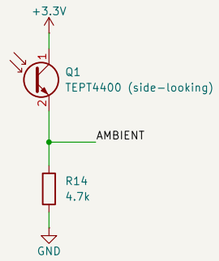


9 · LCD (GC9A01, JST-XH)

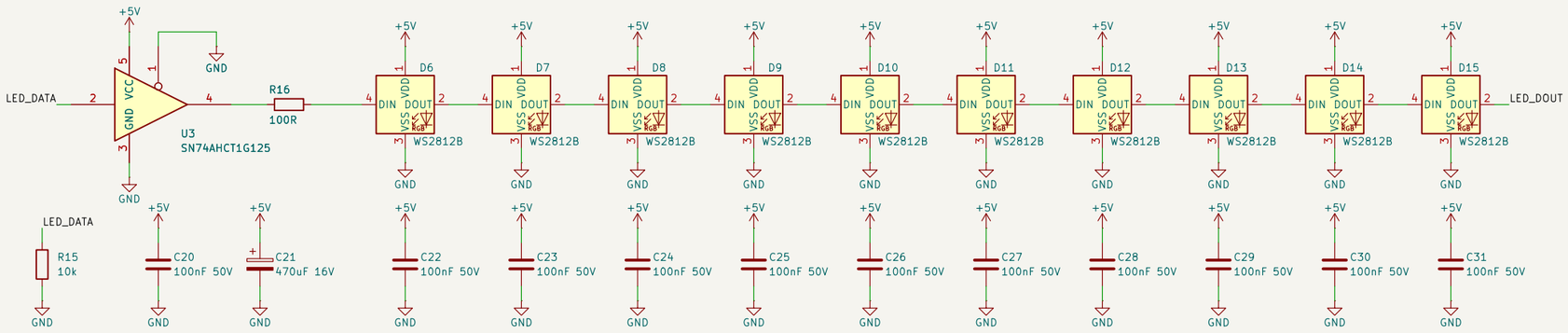
Same order as the module:
BLK CS DC RES SDA SCL VIN GND



10 · AMBIENT LIGHT



11 · LED RING (10 x WS2812B, 5 V data through level shifter)



Hand-soldered build: 0805 passives, socketed modules
ESP32 NodeMCU + TMC2209 + ring knob + LCD + WS2812 ring

Sheet: /	
File: blinds-pcb.kicad_sch	
Title: Smart blinds – control panel	
Size: A3	Date: 2026-11-12
KiCad E.D.A. 10.0.8	Id: 1/1

Rev: 0.1